In [ ]:
```python
import marimo as mo
import altair as alt
import pandas as pd
```

In [ ]:
```python
import duckdb as _ddb
class _Dataset:
    def __init__(self, con):
        self._con = con
    def __call__(self, query):
        _con = self._con
        class _R:
            def df(self): return _con.execute(query).df()
        return _R()
_av_con = _ddb.connect("alpha_vantage.duckdb", read_only=True)
_av_con.execute("SET search_path = 'market_data_20260419013010'")
av_dataset = _Dataset(_av_con)
_gdelt_con = _ddb.connect("gdelt.duckdb", read_only=True)
_gdelt_con.execute("SET search_path = 'gdelt_data'")
gdelt_dataset = _Dataset(_gdelt_con)
```

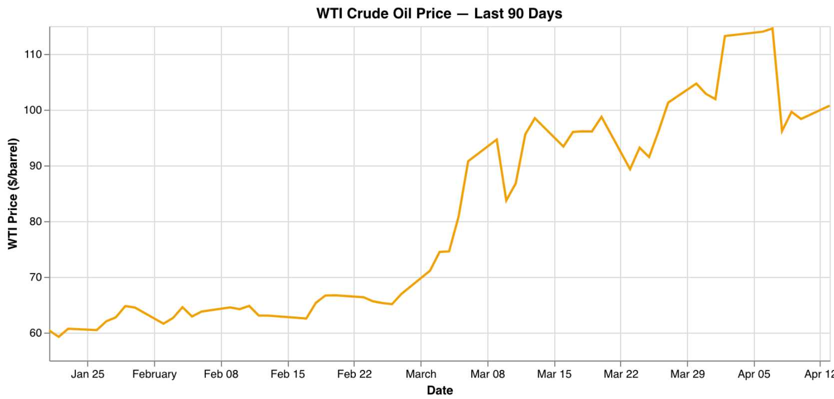
Geopolitical Market Correlation Dashboard

Iran–USA news (GDELT) vs WTI oil and Bitcoin prices (Alpha Vantage)

Market Prices

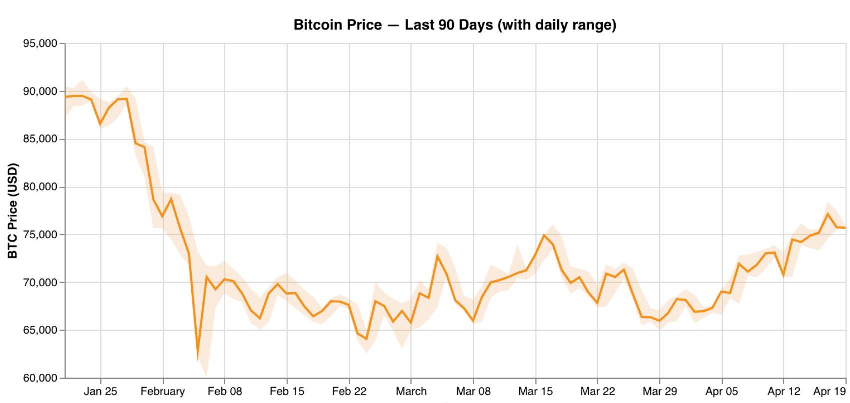
In [ ]:
```python
df_chart1 = av_dataset("""
    SELECT TRY_CAST(date AS DATE) AS date, TRY_CAST(value AS DOUBLE) AS pric
    FROM wti_oil_prices
    WHERE value != '.' AND TRY_CAST(date AS DATE) >= CURRENT_DATE - INTERVAL
    ORDER BY date
""").df()
```

In [ ]:
```python
_chart = alt.Chart(df_chart1).mark_line(color="#f4a100", strokeWidth=2).enco
    x=alt.X("date:T", title="Date"),
    y=alt.Y("price_usd:Q", title="WTI Price ($/barrel)", scale=alt.Scale(zer
    tooltip=[alt.Tooltip("date:T", title="Date"), alt.Tooltip("price_usd:Q",
).properties(title="WTI Crude Oil Price — Last 90 Days", width=700, height=3
_chart
```



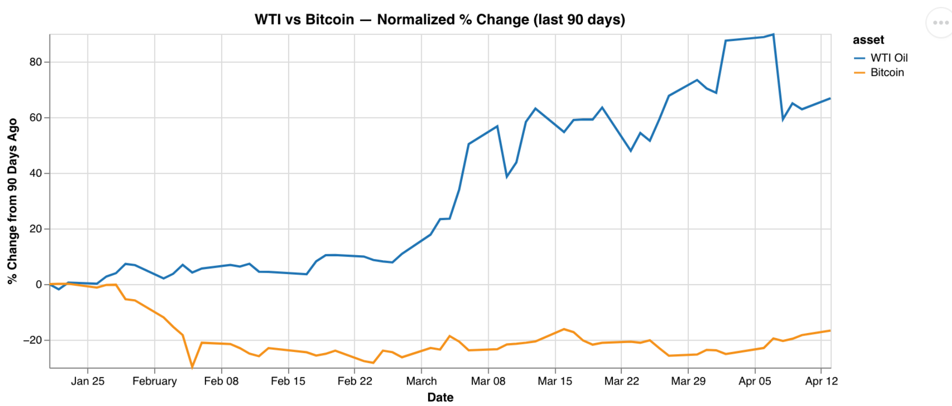
In [ ]:
```
df_chart2 = av_dataset("""
    SELECT TRY_CAST(date AS DATE) AS date, open_usd, high_usd, low_usd, clos
    FROM btc_daily_prices
    WHERE TRY_CAST(date AS DATE) >= CURRENT_DATE - INTERVAL '90 days'
    ORDER BY date
""").df()
```

In [ ]:
```
_band = alt.Chart(df_chart2).mark_area(opacity=0.15, color="#f7931a").encode
    x=alt.X("date:T", title="Date"),
    y=alt.Y("low_usd:Q", scale=alt.Scale(zero=False)),
    y2=alt.Y2("high_usd:Q"),
)
_line = alt.Chart(df_chart2).mark_line(color="#f7931a", strokeWidth=2).encod
    x="date:T",
    y=alt.Y("close_usd:Q", title="BTC Price (USD)", scale=alt.Scale(zero=Fal
    tooltip=[
        alt.Tooltip("date:T", title="Date"),
        alt.Tooltip("high_usd:Q", title="High", format=".0f"),
        alt.Tooltip("low_usd:Q", title="Low", format=".0f"),
        alt.Tooltip("close_usd:Q", title="Close", format=".0f"),
    ],
)
_chart = (_band + _line).properties(
    title="Bitcoin Price — Last 90 Days (with daily range)", width=700, heig
)
_chart
```



```
In [ ]: _combined = df_chart1[["date", "price_usd"]].merge(
            df_chart2[["date", "close_usd"]], on="date", how="inner"
        )
        _combined = _combined.copy()
        _combined["WTI Oil"] = (_combined["price_usd"] / _combined["price_usd"].iloc
        _combined["Bitcoin"] = (_combined["close_usd"] / _combined["close_usd"].iloc
        df_chart3 = _combined[["date", "WTI Oil", "Bitcoin"]].melt(
            "date", var_name="asset", value_name="pct_change"
        )
```

```
In [ ]: _chart = alt.Chart(df_chart3).mark_line(strokeWidth=2).encode(
            x=alt.X("date:T", title="Date"),
            y=alt.Y("pct_change:Q", title="% Change from 90 Days Ago"),
            color=alt.Color(
                "asset:N",
                scale=alt.Scale(domain=["WTI Oil", "Bitcoin"], range=["#1f77b4", "#f
            ),
            tooltip=[alt.Tooltip("date:T"), alt.Tooltip("asset:N"), alt.Tooltip("pct
        ).properties(title="WTI vs Bitcoin — Normalized % Change (last 90 days)", wi
        _chart
```

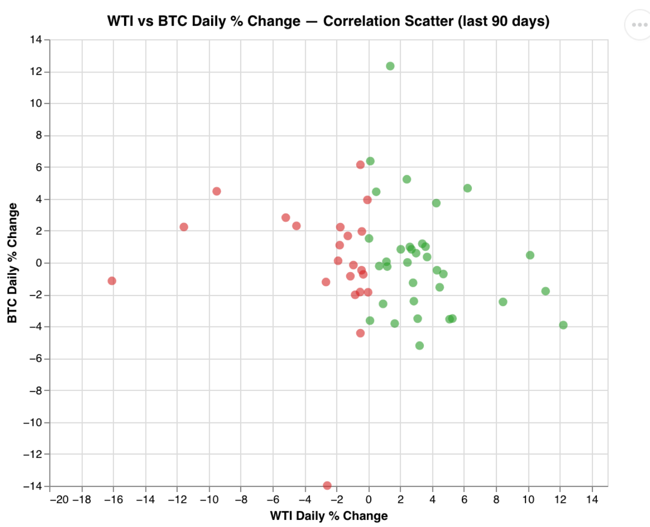


Correlation Analysis

```
In [ ]: df_chart4 = av_dataset("""
            WITH wti_pct AS (
              SELECT TRY_CAST(date AS DATE) AS date, TRY_CAST(value AS DOUBLE) AS pr
                     LAG(TRY_CAST(value AS DOUBLE)) OVER (ORDER BY date) AS prev_pri
              FROM wti_oil_prices WHERE value != '.'
                AND TRY_CAST(date AS DATE) >= CURRENT_DATE - INTERVAL '90 days'
            ),
            btc_pct AS (
              SELECT TRY_CAST(date AS DATE) AS date, close_usd,
                     LAG(close_usd) OVER (ORDER BY date) AS prev_close
              FROM btc_daily_prices
              WHERE TRY_CAST(date AS DATE) >= CURRENT_DATE - INTERVAL '90 days'
            )
            SELECT w.date,
                   ROUND((w.price - w.prev_price) / w.prev_price * 100, 2) AS wti_pc
                   ROUND((b.close_usd - b.prev_close) / b.prev_close * 100, 2) AS bt
            FROM wti_pct w JOIN btc_pct b ON w.date = b.date
```

```
    WHERE w.prev_price IS NOT NULL AND b.prev_close IS NOT NULL
""").df()
```

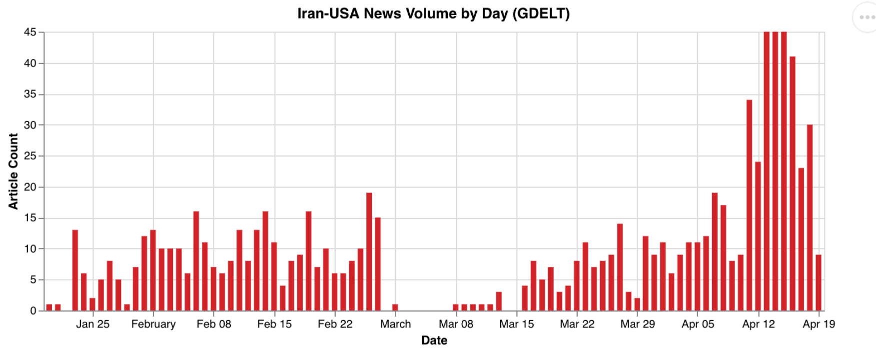
In [ ]:
```
_chart = alt.Chart(df_chart4).mark_circle(size=60, opacity=0.6).encode(
    x=alt.X("wti_pct_change:Q", title="WTI Daily % Change", scale=alt.Scale(
    y=alt.Y("btc_pct_change:Q", title="BTC Daily % Change", scale=alt.Scale(
    color=alt.condition(
        alt.datum.wti_pct_change > 0,
        alt.value("#2ca02c"),
        alt.value("#d62728"),
    ),
    tooltip=[
        alt.Tooltip("date:T", title="Date"),
        alt.Tooltip("wti_pct_change:Q", title="WTI %", format=".2f"),
        alt.Tooltip("btc_pct_change:Q", title="BTC %", format=".2f"),
    ],
).properties(
    title="WTI vs BTC Daily % Change — Correlation Scatter (last 90 days)",
)
_chart
```



Geopolitical News Coverage

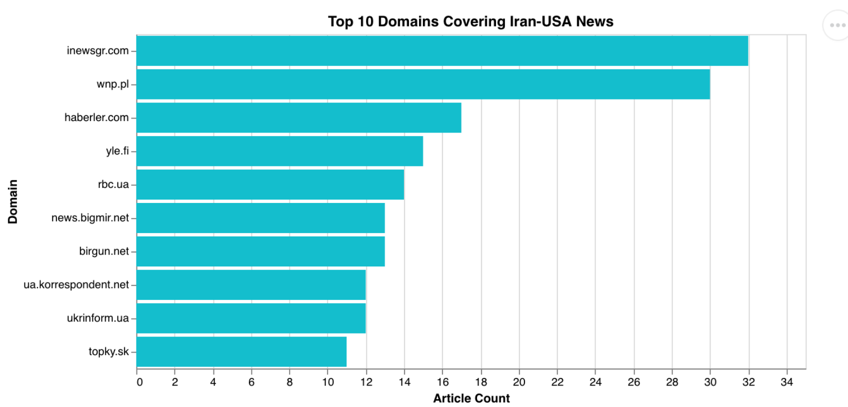
In [ ]:
```
df_chart5 = gdelt_dataset("""
    SELECT DATE_TRUNC('day', seendate)::DATE AS news_date, COUNT(*) AS artic
    FROM articles GROUP BY 1 ORDER BY 1
""").df()
```

In [ ]:
```
_chart = alt.Chart(df_chart5).mark_bar(color="#d62728").encode(
    x=alt.X("news_date:T", title="Date"),
    y=alt.Y("article_count:Q", title="Article Count"),
    tooltip=[alt.Tooltip("news_date:T", title="Date"), alt.Tooltip("article_
).properties(title="Iran-USA News Volume by Day (GDELT)", width=700, height=
_chart
```



In [ ]:
```
df_chart8 = gdelt_dataset("""
    SELECT domain, COUNT(*) AS article_count
    FROM articles WHERE domain IS NOT NULL AND domain != ''
    GROUP BY 1 ORDER BY 2 DESC LIMIT 10
""").df()
```

In [ ]:
```
_chart = alt.Chart(df_chart8).mark_bar(color="#17becf").encode(
    x=alt.X("article_count:Q", title="Article Count"),
    y=alt.Y("domain:N", sort="-x", title="Domain"),
    tooltip=[alt.Tooltip("domain:N", title="Domain"), alt.Tooltip("article_c
).properties(title="Top 10 Domains Covering Iran-USA News", width=600, heigh
_chart
```



Top Headlines on Peak News Days

In [ ]:
```
df_chart16 = gdelt_dataset("""
    WITH top_days AS (
        SELECT DATE_TRUNC('day', seendate)::DATE AS day
        FROM articles
        GROUP BY 1
        ORDER BY COUNT(*) DESC
        LIMIT 5
    )
    SELECT
        td.day AS date,
        a.title,
        a.domain
    FROM top_days td
    JOIN articles a ON DATE_TRUNC('day', a.seendate)::DATE = td.day
```

```
    QUALIFY ROW_NUMBER() OVER (PARTITION BY td.day ORDER BY a.seendate) <= 5
    ORDER BY td.day DESC, a.seendate
""").df()
```

In [ ]:
```
_table = mo.ui.table(df_chart16)
_table
```

Search...

date datetime64[us] Apr 11, 2026 -Apr 16, 2026	title str unique: 24	domain str unique: 23
2026-04-16 00:00:00	ΗΠΑ : Διαψεύδουν αίτημα για παράταση εκεχειρί...	offsite.com.cy
2026-04-16 00:00:00	EEUU : Senado rechaza limitar poderes guerra de Trum...	almomento.net
2026-04-16 00:00:00	美国针对伊朗石油运输网络出台新制裁措施 - 新华网	news.cn
2026-04-16 00:00:00	США оголосили нові санкції проти Ірану	unian.ua
2026-04-16 00:00:00	美國針對伊朗石油運輸網絡出台新製裁措施	portal.sina.com.hk
2026-04-15 00:00:00	EEUU mantiene máxima presión sobre Teherán , no ext...	diariolasamericas.com
2026-04-15 00:00:00	USA : Media : ponad 20 statków handlowych przepłynę...	wnp.pl
2026-04-15 00:00:00	美国副总统称对当前美伊局势进展 感到乐观 |外交|...	163.com
2026-04-15 00:00:00	Suudilerden ABDye ablukayı kaldır baskısı	ntv.com.tr
2026-04-15 00:00:00	USA : Vance : jeśli Iran zrezygnuje z broni jądrowej , jeg...	wnp.pl

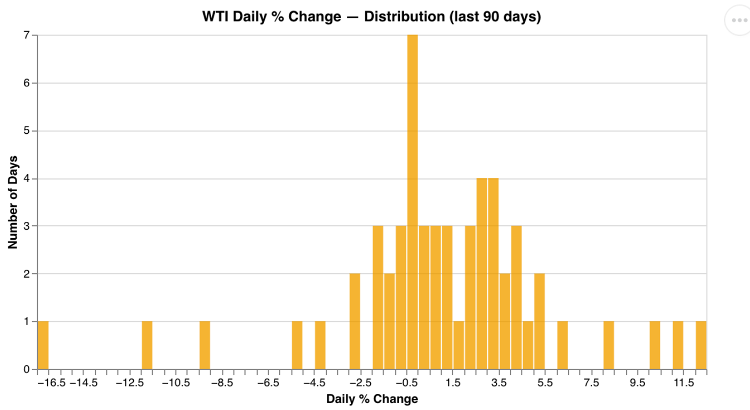
25 rows, 3 columns

No selection

Market Volatility

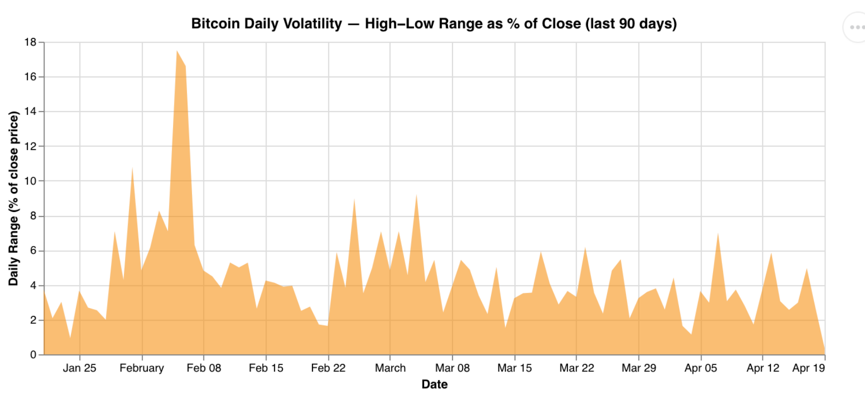
In [ ]:
```
df_chart9 = av_dataset("""
    WITH lagged AS (
      SELECT TRY_CAST(date AS DATE) AS date, TRY_CAST(value AS DOUBLE) AS pr
             LAG(TRY_CAST(value AS DOUBLE)) OVER (ORDER BY date) AS prev_pri
      FROM wti_oil_prices WHERE value != '.'
        AND TRY_CAST(date AS DATE) >= CURRENT_DATE - INTERVAL '90 days'
    )
    SELECT date, ROUND((price - prev_price) / prev_price * 100, 2) AS pct_ch
    FROM lagged WHERE prev_price IS NOT NULL
""").df()
```

In [ ]:
```
_chart = alt.Chart(df_chart9).mark_bar(color="#f4a100", opacity=0.8).encode(
    x=alt.X("pct_change:Q", bin=alt.Bin(step=0.5), title="Daily % Change"),
    y=alt.Y("count():Q", title="Number of Days"),
    tooltip=[alt.Tooltip("pct_change:Q", bin=alt.Bin(step=0.5)), alt.Tooltip
).properties(title="WTI Daily % Change - Distribution (last 90 days)", width
_chart
```



In [ ]:
```
df_chart10 = av_dataset("""
    SELECT TRY_CAST(date AS DATE) AS date,
           ROUND((high_usd - low_usd) / close_usd * 100, 2) AS range_pct
    FROM btc_daily_prices
    WHERE TRY_CAST(date AS DATE) >= CURRENT_DATE - INTERVAL '90 days'
    ORDER BY date
""").df()
```

In [ ]:
```
_chart = alt.Chart(df_chart10).mark_area(color="#f7931a", opacity=0.6).encod
    x=alt.X("date:T", title="Date"),
    y=alt.Y("range_pct:Q", title="Daily Range (% of close price)"),
    tooltip=[alt.Tooltip("date:T", title="Date"), alt.Tooltip("range_pct:Q",
).properties(
    title="Bitcoin Daily Volatility — High-Low Range as % of Close (last 90
)
_chart
```

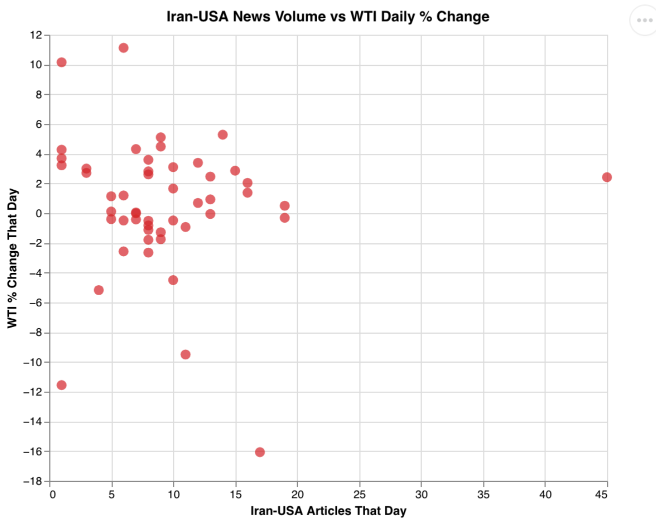


News–Market Correlation

In [ ]:
```
_news = df_chart5.rename(columns={"news_date": "date"})
df_chart11 = _news.merge(df_chart9, on="date", how="inner")
```

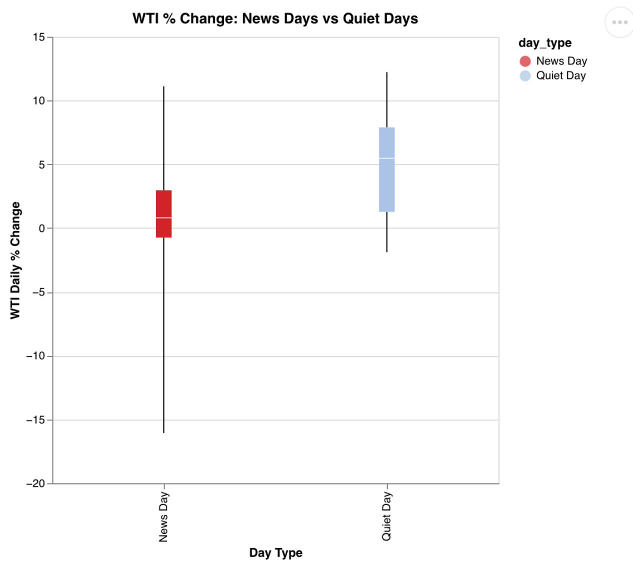
In [ ]:
```
_chart = alt.Chart(df_chart11).mark_circle(size=80, color="#d62728", opacity
    x=alt.X("article_count:Q", title="Iran-USA Articles That Day"),
    y=alt.Y("pct_change:Q", title="WTI % Change That Day"),
    tooltip=[
        alt.Tooltip("date:T", title="Date"),
```

```
        alt.Tooltip("article_count:Q", title="Articles"),
        alt.Tooltip("pct_change:Q", title="WTI % Change", format=".2f"),
    ],
).properties(title="Iran-USA News Volume vs WTI Daily % Change", width=500,
_chart
```



In [ ]:
```
_news_dates = set(df_chart5["news_date"].astype(str))
df_chart12 = df_chart9.copy()
df_chart12["day_type"] = df_chart12["date"].astype(str).apply(
    lambda d: "News Day" if d in _news_dates else "Quiet Day"
)
```

In [ ]:
```
_chart = alt.Chart(df_chart12).mark_boxplot(extent="min-max").encode(
    x=alt.X("day_type:N", title="Day Type"),
    y=alt.Y("pct_change:Q", title="WTI Daily % Change"),
    color=alt.Color(
        "day_type:N",
        scale=alt.Scale(domain=["News Day", "Quiet Day"], range=["#d62728",
    ),
    tooltip=[alt.Tooltip("day_type:N"), alt.Tooltip("pct_change:Q", format="
).properties(title="WTI % Change: News Days vs Quiet Days", width=400, heigh
_chart
```

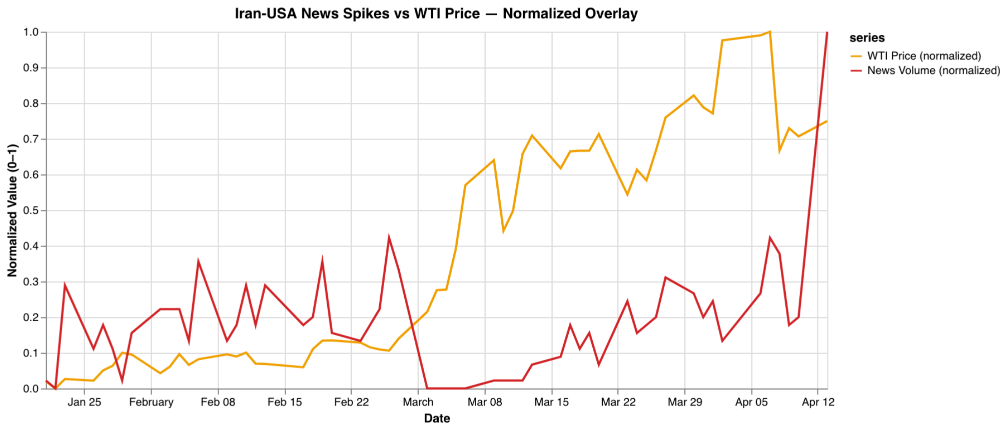


In [ ]:

```
_news = df_chart5.rename(columns={"news_date": "date"})
_lag = df_chart1.merge(_news, on="date", how="left")
_lag["article_count"] = _lag["article_count"].fillna(0)
_lag["wti_norm"] = (
    (_lag["price_usd"] - _lag["price_usd"].min())
    / (_lag["price_usd"].max() - _lag["price_usd"].min())
)
_lag["news_norm"] = (
    (_lag["article_count"] - _lag["article_count"].min())
    / (_lag["article_count"].max() - _lag["article_count"].min() + 1e-9)
)
df_chart13 = _lag[["date", "wti_norm", "news_norm"]].melt(
    "date", var_name="series", value_name="value"
)
df_chart13 = df_chart13.assign(
    series=df_chart13["series"].map(
        {"wti_norm": "WTI Price (normalized)", "news_norm": "News Volume (nc
    )
)
```

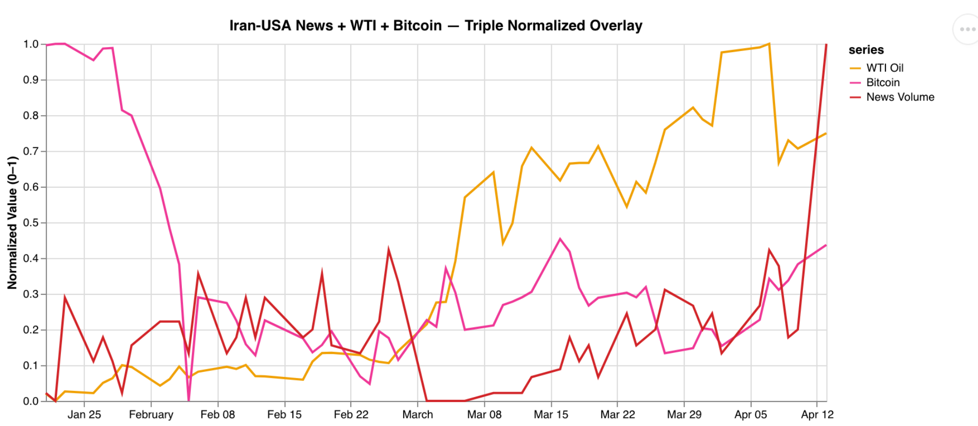
In [ ]:

```
_chart = alt.Chart(df_chart13).mark_line(strokeWidth=2).encode(
    x=alt.X("date:T", title="Date"),
    y=alt.Y("value:Q", title="Normalized Value (0–1)"),
    color=alt.Color(
        "series:N",
        scale=alt.Scale(
            domain=["WTI Price (normalized)", "News Volume (normalized)"],
            range=["#f4a100", "#d62728"],
        ),
    ),
    tooltip=[alt.Tooltip("date:T"), alt.Tooltip("series:N"), alt.Tooltip("va
).properties(
    title="Iran–USA News Spikes vs WTI Price — Normalized Overlay", width=70
)
_chart
```



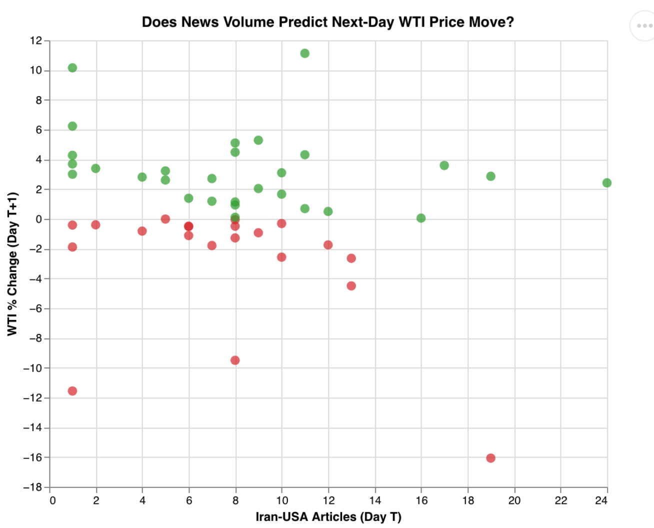
In [ ]:
```
_wti = df_chart1[["date", "price_usd"]].copy()
_btc = df_chart2[["date", "close_usd"]].copy()
_news = df_chart5.rename(columns={"news_date": "date"}).copy()
_m = _wti.merge(_btc, on="date", how="inner").merge(_news, on="date", how="l
_m["article_count"] = _m["article_count"].fillna(0)
for _col, _label in [("price_usd", "WTI Oil"), ("close_usd", "Bitcoin"), ("a
    _mn, _mx = _m[_col].min(), _m[_col].max()
    _m[_label] = (_m[_col] - _mn) / (_mx - _mn + 1e-9)
df_chart14 = _m[["date", "WTI Oil", "Bitcoin", "News Volume"]].melt(
    "date", var_name="series", value_name="value"
)
```

In [ ]:
```
_chart = alt.Chart(df_chart14).mark_line(strokeWidth=2).encode(
    x=alt.X("date:T", title="Date"),
    y=alt.Y("value:Q", title="Normalized Value (0–1)"),
    color=alt.Color(
        "series:N",
        scale=alt.Scale(
            domain=["WTI Oil", "Bitcoin", "News Volume"],
            range=["#f4a101", "#f2399a", "#d62728"],
        ),
    ),
    tooltip=[alt.Tooltip("date:T"), alt.Tooltip("series:N"), alt.Tooltip("va
).properties(title="Iran-USA News + WTI + Bitcoin — Triple Normalized Overla
_chart
```



```
In [ ]: import pandas as _pd
        _news = df_chart5.rename(columns={"news_date": "date"}).copy()
        _news["date"] = _pd.to_datetime(_news["date"])
        _wti = df_chart9.copy()
        _wti["date"] = _pd.to_datetime(_wti["date"])
        _wti["prev_date"] = _wti["date"] - _pd.Timedelta(days=1)
        df_chart15 = _news.merge(
            _wti[["prev_date", "pct_change"]],
            left_on="date", right_on="prev_date",
            how="inner",
        )[["date", "article_count", "pct_change"]].rename(columns={"pct_change": "ne
```

```
In [ ]: _chart = alt.Chart(df_chart15).mark_circle(size=70, opacity=0.7).encode(
            x=alt.X("article_count:Q", title="Iran-USA Articles (Day T)"),
            y=alt.Y("next_day_wti_pct:Q", title="WTI % Change (Day T+1)", scale=alt.
            color=alt.condition(
                alt.datum.next_day_wti_pct > 0,
                alt.value("#2ca02c"),
                alt.value("#d62728"),
            ),
            tooltip=[
                alt.Tooltip("date:T", title="Date"),
                alt.Tooltip("article_count:Q", title="Articles"),
                alt.Tooltip("next_day_wti_pct:Q", title="Next-day WTI %", format=".2
            ],
        ).properties(title="Does News Volume Predict Next-Day WTI Price Move?", widt
        _chart
```



Events Deep Dive

```
In [ ]: df_chart17_raw = av_dataset("""
            SELECT
                TRY_CAST(w.date AS DATE) AS date,
                TRY_CAST(w.value AS DOUBLE) AS wti_price,
                b.close_usd AS btc_close
            FROM wti_oil_prices w
```

```
    JOIN btc_daily_prices b ON TRY_CAST(w.date AS DATE) = TRY_CAST(b.date AS
    WHERE w.value != '.'
      AND TRY_CAST(w.date AS DATE) BETWEEN '2026-04-05' AND '2026-04-19'
    ORDER BY date
""").df()
_base_wti = df_chart17_raw["wti_price"].iloc[0]
_base_btc = df_chart17_raw["btc_close"].iloc[0]
df_chart17_raw["WTI Oil"] = (df_chart17_raw["wti_price"] / _base_wti - 1) *
df_chart17_raw["Bitcoin"] = (df_chart17_raw["btc_close"] / _base_btc - 1) *
df_chart17 = df_chart17_raw[["date", "WTI Oil", "Bitcoin"]].melt(
    "date", var_name="asset", value_name="pct_from_apr5"
)
```

In [ ]:
```
_chart = alt.Chart(df_chart17).mark_line(strokeWidth=2.5).encode(
    x=alt.X("date:T", title="Date"),
    y=alt.Y("pct_from_apr5:Q", title="% Change from Apr 5"),
    color=alt.Color(
        "asset:N",
        scale=alt.Scale(domain=["WTI Oil", "Bitcoin"], range=["#1f77b4", "#f
    ),
    tooltip=[
        alt.Tooltip("date:T"),
        alt.Tooltip("asset:N"),
        alt.Tooltip("pct_from_apr5:Q", format=".1f", title="% from Apr 5"),
    ],
).properties(title="WTI vs BTC During Strait of Hormuz Threat (Apr 5–19)", w
_chart
```